

Synopsis

TITLE	IntelliCV: A Smart Resume Builder	
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1. Introduction

In today's competitive job market, resumes are crucial for securing interviews, yet most are generic and not aligned with job requirements. *IntelliCV* is a smart, AI-based web platform that automates personalized resume creation using Artificial Intelligence (AI) and Natural Language Processing (NLP). The system analyzes the user's academic and professional documents—projects, certificates, and achievements—to generate an optimized resume tailored to a given job description (JD). By applying semantic similarity techniques and Sentence Transformers, IntelliCV enhances job relevance and candidate–role matching.

2. Existing System

Conventional resume builders depend on static templates and manual data entry. They:

- Lack semantic understanding of job descriptions.
- Do not extract skills or relevant details from uploaded files.
- Store all data in relational databases, limiting flexibility for unstructured content like PDFs or certificates.

These limitations result in generic resumes that fail to highlight a candidate's most relevant skills and experiences.

3. Proposed System

IntelliCV introduces an AI-driven platform that bridges this gap by combining relational and NoSQL data management with NLP-powered content analysis.

Key features include:

- **Automated Document Parsing:** Extraction of text from uploaded files (PDF/DOCX).
- **Skill Extraction:** Identification of technical and soft skills using NLP-based Named Entity Recognition (NER).
- **Semantic Matching:** Computation of similarity between job descriptions and user portfolio data using BERT-based sentence embeddings.
- **Dynamic Resume Generation:** Automatic assembly of the most relevant experiences and skills into a structured resume format.
- **Hybrid Storage Architecture:** SQL for structured data and MongoDB for embeddings and logs.

4. Relational Database Structure

The relational data is managed using **MySQL** with six interrelated tables:

- **User** (*user_id, name, email, contact, profile_summary*)
- **Document** (*doc_id, user_id, file_name, file_type, upload_date, text_extracted*)
- **Project** (*proj_id, user_id, title, description, tech_stack, duration*)
- **Certificate** (*cert_id, user_id, title, issuing_org, issue_date, file_path*)
- **Job_Description** (*job_id, user_id, title, company, jd_text, embedding_vector*)
- **Generated_Resume** (*resume_id, user_id, job_id, generated_text, match_score, timestamp*)

Key Relationships:

- *One-to-many:* User → Document, Project, Certificate
- *One-to-one:* Job_Description → Generated_Resume

5. RDBMS AND NoSQL Integration

While structured data (users, projects, certificates) is stored in MySQL, **MongoDB** complements it by handling unstructured and high-dimensional data:

- **embeddings collection:** Stores vector embeddings of user documents and job descriptions for semantic similarity computations.
- **logs collection:** Maintains feedback data, model performance, and usage analytics for continuous improvement.

6. Societal Concern

IntelliCV addresses the societal problem of unemployment and skill mismatch by enabling smarter job applications. By harnessing AI, NLP, and ML, the platform ensures fairer candidate evaluation and improved employability. It supports inclusive hiring by providing all job seekers—especially students—with tools to present their qualifications effectively.